Universidad Carlos III de Madrid

Bachelor in Aerospace Engineering

Mechanics Applied to Aerospace Engineering

Particle Connected to a Spool

Laboratory Session 1

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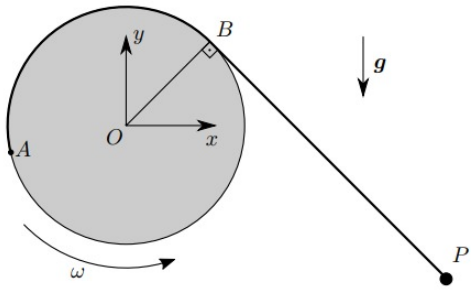
1 Introduction and Problem Statement

In this laboratory report, we will investigate the dynamics of a particle P of mass m attached to a massless string of total length l, rolled around a rotating cylinder of radius a. The system gives us a moving tangency point $B(t)$ between the string and the spool and a non-rolled string length $\xi(t)$ that depends on time. As we will show later, the motion equations of the system can be expressed in function of the only variable ϕ, which means that the problem only has 1 Configuration Degree of Freedom, the angle between the horizontal and the line connecting the center of the cylinder with the tangency point B.

Calculating the kinematics of the problem in a polar reference system centered at the center of the cylinder and rotating with angular velocity $\dot{\phi}$ and applying Newton's Second Law taking into account the gravity and tension forces, we can derive a differential equation that does not depend on the tension for ϕ. In Matlab, we have used function `ode45` to numerically solve for ϕ in different 5 test cases with 5 different initial conditions. We have used the solution these equations to describe the motion of point P in each case and we have the analyzed the conservation of its mechanical energy and its behaviour in each case.

1.1 Problem Statement

The particle P has mass m and hangs from a massless, inextensible string of total length l. The other end of the string is attached at a point A of a spool of radius a and centre O, which turns counterclockwise at a constant rate ω. B is the point where the straight part of the string leaves the spool, at an angle ϕ from Ox, and ξ is the length of the straight segment BP. The parameters are $g = 9.81\,\mathrm{m/s^2}$, $m = 0.1\,\mathrm{kg}$, $a = 0.2\,\mathrm{m}$, $\phi(0) = 0$ and $\xi(0) = 1\,\mathrm{m}$; the five cases differ in ω and in the initial velocity $\dot{x}_P(0)$ (Table 1).



1.2 Degrees of Freedom and Constraint

In the plane, P needs two coordinates, and the pair (ϕ, ξ) is a valid choice through $\mathbf{r}_P = a\hat{\mathbf{e}}_r - \xi\hat{\mathbf{e}}_\phi$ (Eq. (1.9)). The string removes one of them. The anchor A is at the angle $\theta_A(t) = \theta_{A0} + \omega t$, the wrapped arc BA has length $a(\theta_A - \phi)$, and the total length does not change:

$$a\,(\theta_{A0} + \omega t - \phi) + \xi = l \quad \Longrightarrow \quad \xi(t, \phi) = \xi_0 - a\,(\omega t - \phi), \qquad \xi_0 = l - a\theta_{A0}. \tag{1.1}$$

$$\text{C.D.O.F.} = 2 - 1 = 1, \tag{1.2}$$

$\phi(t)$ gives the constraint for the kinematics when t is known. With the chain rule:

$$\dot{\xi} = \frac{\partial \xi}{\partial t} + \frac{\partial \xi}{\partial \phi}\,\dot{\phi} = a\dot{\phi} - a\omega. \tag{1.3}$$

1.3 Reference Frame and Polar Basis

To derive the kinematic vectors, we define a polar reference system attached to the rotating radius OB:

$$\hat{e}_r = \cos\phi\,\hat{i} + \sin\phi\,\hat{j} \tag{1.4}$$

$$\hat{e}_\phi = -\sin\phi\,\hat{i} + \cos\phi\,\hat{j} \tag{1.5}$$

Their time derivatives are given by:

$$\dot{\hat{e}}_r = \dot{\phi}\hat{e}_\phi, \quad \dot{\hat{e}}_\phi = -\dot{\phi}\hat{e}_r \tag{1.6}$$

The Cartesian unit vectors in terms of the polar basis:

$$\hat{\imath} = \hat{e}_r\cos\phi - \hat{e}_\phi\sin\phi, \quad \hat{\jmath} = \hat{e}_r\sin\phi + \hat{e}_\phi\cos\phi \tag{1.7}$$

$$\dot{\hat{\mathbf{e}}}_r = \frac{d\hat{\mathbf{e}}_r}{d\phi}\,\dot{\phi} = (-\sin\phi\,\hat{\boldsymbol{\imath}} + \cos\phi\,\hat{\boldsymbol{\jmath}})\,\dot{\phi} = \dot{\phi}\,\hat{\mathbf{e}}_\phi, \qquad \dot{\hat{\mathbf{e}}}_\phi = \frac{d\hat{\mathbf{e}}_\phi}{d\phi}\,\dot{\phi} = -(\cos\phi\,\hat{\boldsymbol{\imath}} + \sin\phi\,\hat{\boldsymbol{\jmath}})\,\dot{\phi} = -\dot{\phi}\,\hat{\mathbf{e}}_r. \tag{1.8}$$

1.4 Position, Velocity and Acceleration of P

B lies on the spool, $\mathbf{r}_B = a\hat{\mathbf{e}}_r$, and the string leaves it tangentially along $-\hat{\mathbf{e}}_\phi$:

$$\mathbf{r}_P = a\hat{\mathbf{e}}_r - \xi\hat{\mathbf{e}}_\phi \quad\Longleftrightarrow\quad x_P = a\cos\phi + \xi\sin\phi, \qquad y_P = a\sin\phi - \xi\cos\phi. \tag{1.9}$$

Differentiating with the product rule and using Eqs. (1.3) and (1.8):

$$\mathbf{v}_P = a\dot{\hat{\mathbf{e}}}_r - \dot{\xi}\,\hat{\mathbf{e}}_\phi - \xi\dot{\hat{\mathbf{e}}}_\phi = a\dot{\phi}\,\hat{\mathbf{e}}_\phi - (a\dot{\phi} - a\omega)\,\hat{\mathbf{e}}_\phi + \xi\dot{\phi}\,\hat{\mathbf{e}}_r \quad\Longrightarrow\quad \boxed{\mathbf{v}_P = \xi\dot{\phi}\,\hat{\mathbf{e}}_r + a\omega\,\hat{\mathbf{e}}_\phi}. \tag{1.10}$$

Differentiating again, with a and ω constant:

$$\mathbf{a}_P = (\dot{\xi}\dot{\phi} + \xi\ddot{\phi})\,\hat{\mathbf{e}}_r + \xi\dot{\phi}\,\dot{\hat{\mathbf{e}}}_r + a\omega\,\dot{\hat{\mathbf{e}}}_\phi = \left[(a\dot{\phi} - a\omega)\dot{\phi} + \xi\ddot{\phi} - a\omega\dot{\phi}\right]\hat{\mathbf{e}}_r + \xi\dot{\phi}^2\,\hat{\mathbf{e}}_\phi$$

$$\Longrightarrow \quad \boxed{\mathbf{a}_P = \left(a\dot{\phi}^2 + \xi\ddot{\phi} - 2a\omega\dot{\phi}\right)\hat{\mathbf{e}}_r + \xi\dot{\phi}^2\,\hat{\mathbf{e}}_\phi} \tag{1.11}$$

2 Mathematical Model and Dynamical Formulation

2.1 Forces on the Particle

Two forces act on P: the tension, which pulls it towards B ($T \geq 0$, since a string cannot push), and its weight:

$$\mathbf{T} = T\,\hat{\mathbf{e}}_\phi; \qquad m\mathbf{g} = -mg\,\hat{\boldsymbol{\jmath}} = -mg\,(\sin\phi\,\hat{\mathbf{e}}_r + \cos\phi\,\hat{\mathbf{e}}_\phi)\,. \tag{2.1}$$

2.2 Equation of Motion and String Tension

Newton's second law in the inertial frame, $m\mathbf{a}_P = \mathbf{T} + m\mathbf{g}$, projected on $\hat{\mathbf{e}}_r$ and $\hat{\mathbf{e}}_\phi$ with Eqs. (1.11) and (2.1):

$$\hat{\mathbf{e}}_r: \quad m\left(a\dot{\phi}^2 + \xi\ddot{\phi} - 2a\omega\dot{\phi}\right) = -mg\sin\phi, \tag{2.2}$$

$$\hat{\mathbf{e}}_\phi: \quad m\xi\dot{\phi}^2 = T - mg\cos\phi. \tag{2.3}$$

T only appears in Eq. (2.3). Dividing Eq. (2.2) by $m\xi$ (valid while $\xi > 0$) and substituting Eq. (1.1):

$$\boxed{\ddot{\phi} = \frac{-g\sin\phi - a\dot{\phi}^2 + 2a\omega\dot{\phi}}{\xi_0 - a(\omega t - \phi)}} \qquad \boxed{T(t,\phi,\dot{\phi}) = m\,\xi(t,\phi)\,\dot{\phi}^2 + mg\cos\phi} \tag{2.4}$$

The equation is non-linear and depends on t explicitly through $\xi(t,\phi)$.

3 Numerical Solution in Matlab

With $\mathbf{X} = [X_1, X_2]^\mathsf{T} = [\phi, \dot{\phi}]^\mathsf{T}$, Eq. (2.4) becomes

$$\frac{d\mathbf{X}}{dt} = \mathbf{f}(t, \mathbf{X}) = \begin{pmatrix} X_2 \\ \dfrac{-g \sin X_1 - a X_2^2 + 2a\omega X_2}{\xi_0 - a(\omega t - X_1)} \end{pmatrix}, \tag{3.1}$$

which is the local function `diffeq(t,X,g,omega,a,xi0)` of `main.m`. It is integrated with `ode45` with `RelTol` $= 10^{-8}$ and a maximum step of 10 ms. The data give $\dot{x}_P(0)$, not $\dot{\phi}(0)$. At $\phi_0 = 0$ we have $\hat{\mathbf{e}}_r = \hat{\boldsymbol{\imath}}$ and $\hat{\mathbf{e}}_\phi = \hat{\boldsymbol{\jmath}}$, so Eq. (1.10) gives $\dot{x}_P(0) = \xi_0 \dot{\phi}_0$ and $\dot{\phi}_0 = \dot{x}_P(0)/\xi_0$; the vertical component $\dot{y}_P(0) = a\omega$ depends on the spool's angular velocity.

`stopfun` returns `value` $= [\xi - \xi_{\text{tol}}, T]$, `isterminal` $= [1, 1]$ and `direction` $= [-1, -1]$, with ξ and T from Eqs. (1.1) and (2.4). Near the spool $\ddot{\phi} \propto 1/\xi$, and in our first tries with $\xi_{\text{tol}} = 0$, `ode45` stopped before reaching $\xi = 0$ without reporting the event. The contact event is therefore placed at $\xi_{\text{tol}} = 10^{-6}$ m, smaller than any real string thickness.

4 Simulation Results and Trajectories

Figs. 1–5 show, for each case, $\phi(t)$ and $\xi(t)$ (a), the tension $T(t)$ (b), the trajectory of P with the spool (c) and the mechanical energy $E(t)$ (d). Case 2 never triggers an event, so it is integrated over five periods. Table 1 summarises the five cases at the end of the section.

Case 1. With $\phi = \dot{\phi} = 0$ the numerator of Eq. (3.1) is zero for every t, so $\phi \equiv 0$ is an exact solution. $\xi = \xi_0 - a\omega t$ drops linearly so the spool acts as if it was a winch, P rises vertically at $a\omega = 0.02$ m/s, and since it does not accelerate, $T = mg = 0.981$ N (Fig. 1). P reaches the spool at $t = \xi_0/(a\omega) = 50$ s (the event stops the run at 49.999 95 s), at $B = (a, 0)$; the numerical solution keeps $\phi = 0$.

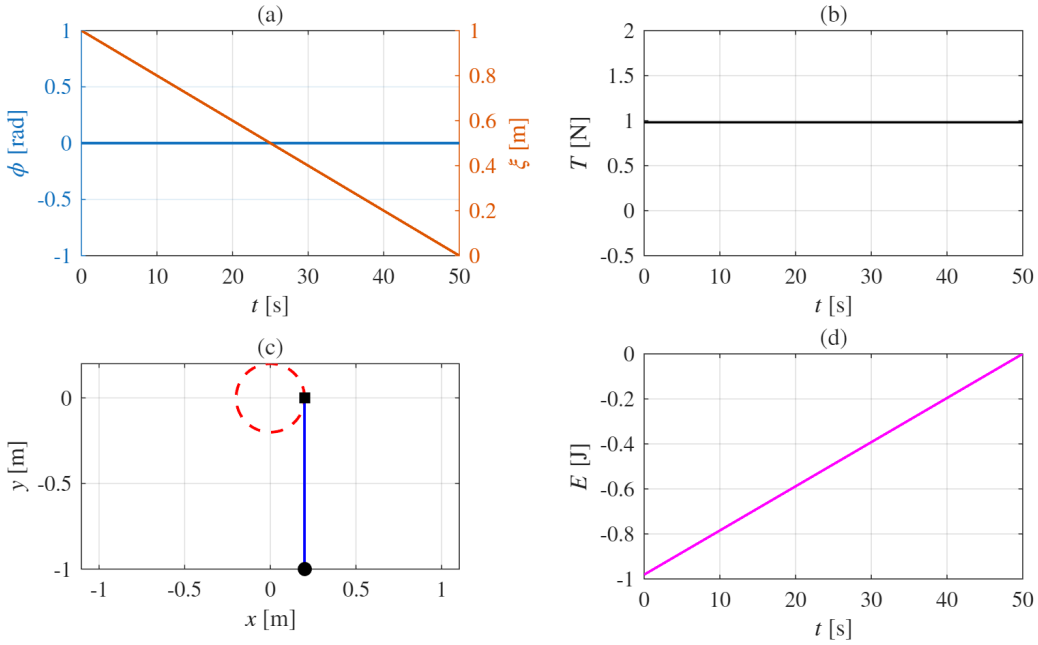


Figure 1: Case 1 ($\omega = 0.1$ rad/s, $\dot{x}_P(0) = 0$). (a) ϕ and ξ; (b) tension; (c) trajectory and spool (• start, ■ end); (d) energy.

Case 2. With the spool at rest, $\xi = \xi_0 + a\phi$ is not constant: it oscillates between 0.934 41

and 1.062 82 m as the string wraps and unwraps (Fig. 2). The swing is not symmetric. ϕ goes from −0.327 95 rad to 0.314 09 rad, because both turning points are at the same height (energy is conserved) while the pivot B moves with ϕ. The tension stays between 0.929 and 1.081 N, so no event occurs and the motion is periodic.

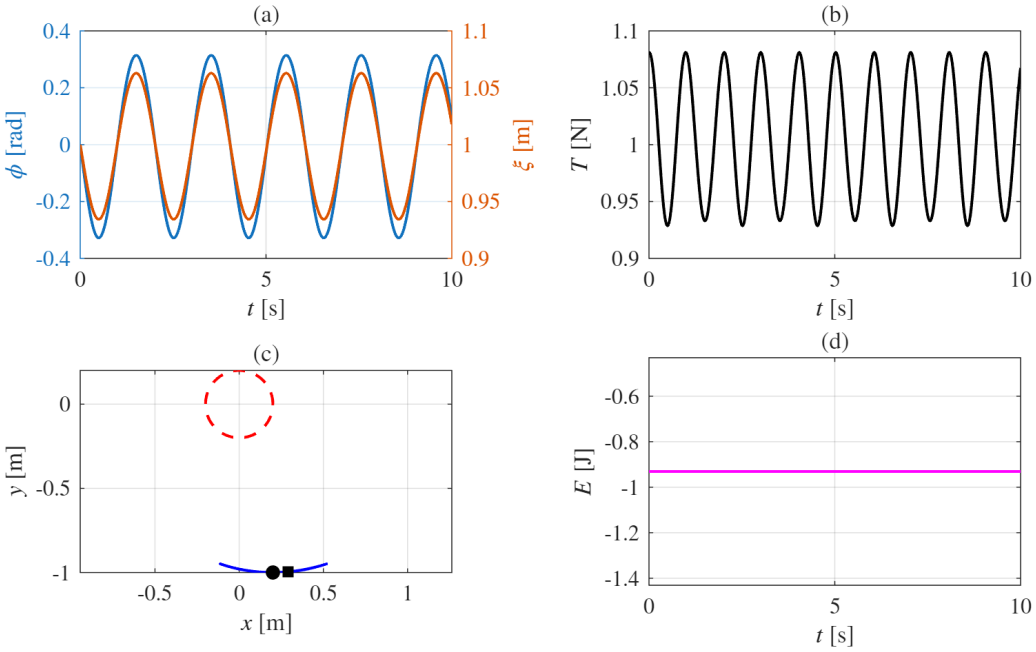


Figure 2: Case 2 ($\omega = 0$, $\dot{x}_P(0) = -1\,\mathrm{m/s}$). Panels and markers as in Fig. 1.

Case 3. The fast launch to the left wraps the string clockwise around the spool. ϕ decreases to −5 rad (−286.5°), where $\xi = \xi_0 + a\phi = 0$, and the trajectory follows a spiral path in (Fig. 3c). The tension starts at $m\dot{x}_P(0)^2/\xi_0 + mg = 10.98\,\mathrm{N}$ and grows without bound before contact (Fig. 3b).

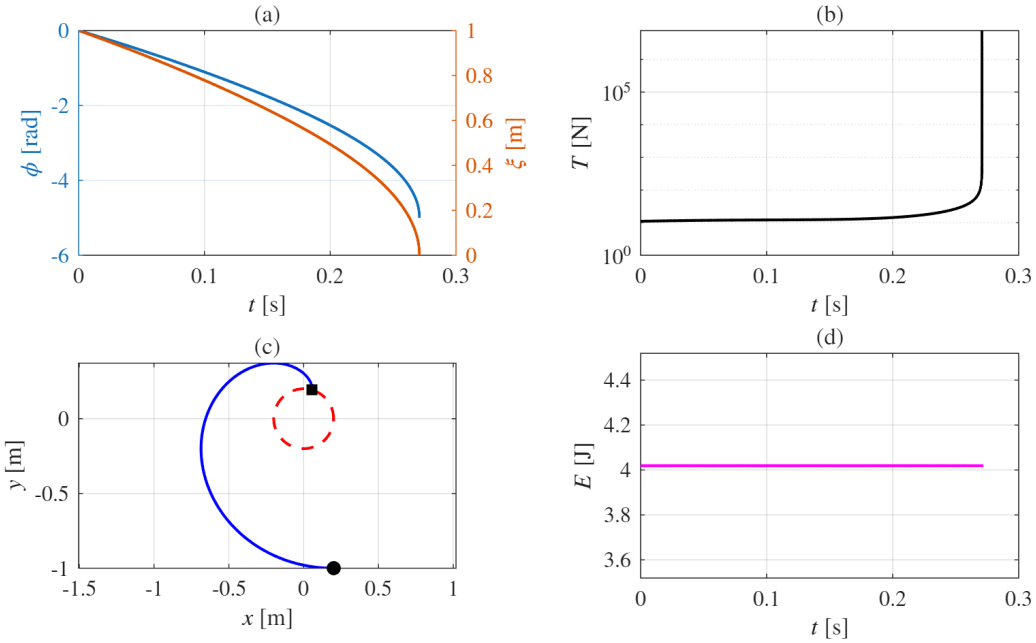


Figure 3: Case 3 ($\omega = 0$, $\dot{x}_P(0) = -10\,\mathrm{m/s}$). With $\omega = 0$, $\xi = \xi_0 + a\phi$, so the two curves of (a) have the same shape. (b) is plotted in logarithmic scale because the tension approaches ∞ as ξ approaches 0. Panels and markers as in Fig. 1.

Case 4. P first swings to the right, unwinds string up to $\xi = 1.2533\,\mathrm{m}$ and turns back at $\phi_{\max} = 1.266\,67\,\mathrm{rad}$. It then passes the lowest point and rises on the left, where part of the string is wrapped, until the tension vanishes at $t = 1.693\,47\,\mathrm{s}$ (Fig. 4).

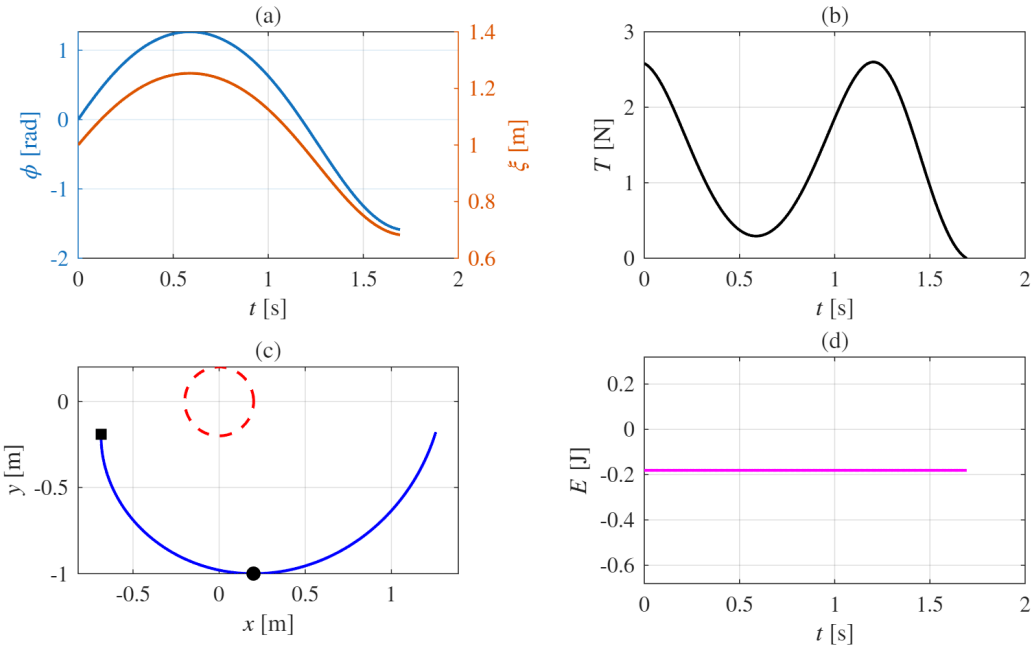


Figure 4: Case 4 ($\omega = 0$, $\dot{x}_P(0) = 4\,\mathrm{m/s}$). Panels and markers as in Fig. 1.

Case 5. This case adds the oscillation of case 2 to the winding of case 1. As the string gets shorter the oscillation speeds up and its amplitude grows: the swings to the right go from 0.3164 rad to 0.7006 rad, and those to the left from −0.3360 rad to −1.0880 rad. The tension oscillates around a slowly rising mean staying positive (Fig. 5b).

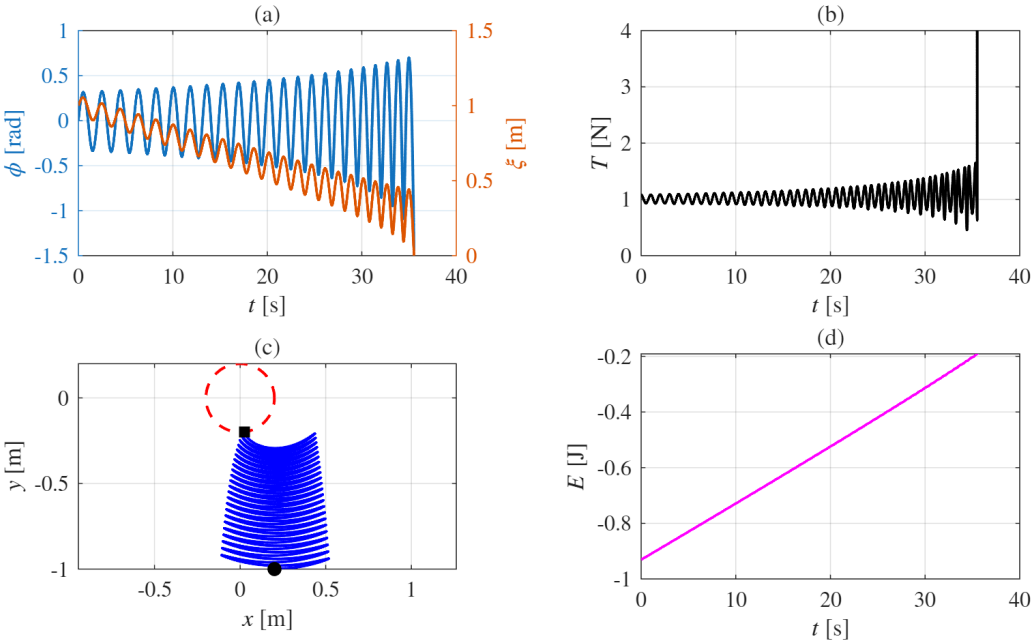


Figure 5: Case 5 ($\omega = 0.1\,\mathrm{rad/s}$, $\dot{x}_P(0) = 1\,\mathrm{m/s}$). (b) is cut at 4 N to be able to observe the oscillations previous to the singularity. Panels and markers as in Fig. 1.

Table 1: Study cases and main results. t_{end}: time of the event (final time in case 2). Event $\xi = 0$: the particle reaches the spool; $T = 0$: the string goes slack.

Case	ω (rad/s)	$\dot{x}_P(0)$ (m/s)	E_0 (J)	t_{end} (s)	Event	$x_P(t_{\text{end}})$ (m)	$y_P(t_{\text{end}})$ (m)	$E(t_{\text{end}})$ (J)
1	0.1	0	−0.980 98	49.999 95	$\xi = 0$	0.200 00	0.000 00	0.000 02
2	0	−1	−0.931 00	10.000 00	none	0.290 41	−0.995 93	−0.931 00
3	0	−10	4.019 00	0.270 95	$\xi = 0$	0.056 73	0.191 78	4.018 99
4	0	4	−0.181 00	1.693 47	$T = 0$	−0.685 76	−0.189 66	−0.181 00
5	0.1	1	−0.930 98	35.509 42	$\xi = 0$	0.024 29	−0.198 52	−0.191 20

5 Physical Discussion and Energy

5.1 Mechanical Energy

The mechanical energy of P, with $V = mgy_P$ and $|\mathbf{v}_P|^2 = \xi^2\dot{\phi}^2 + a^2\omega^2$ from Eq. (1.10), is

$$E = \tfrac{1}{2}m\left(\xi^2\dot{\phi}^2 + a^2\omega^2\right) + mg\,(a\sin\phi - \xi\cos\phi)\,. \tag{5.1}$$

By the work-energy theorem $dK/dt = (\mathbf{T} + m\mathbf{g})\cdot\mathbf{v}_P$, (the weight is conservative) $m\mathbf{g}\cdot\mathbf{v}_P = -dV/dt$. Only the power of the tension is left:

$$\frac{dE}{dt} = \mathbf{T}\cdot\mathbf{v}_P = T\,\hat{\mathbf{e}}_\phi\cdot\left(\xi\dot{\phi}\,\hat{\mathbf{e}}_r + a\omega\,\hat{\mathbf{e}}_\phi\right) = Ta\omega. \tag{5.2}$$

While the string is in tension $T > 0$, E is conserved if and only if $\omega = 0$ (cases 2–4). For $\omega > 0$ (cases 1 and 5) E increases all the time, this is because the spool winds the string in at speed $a\omega$ against the tension, and that work goes into P.

Panel (d) of each case figure agrees with Eq. (5.2). In cases 2–4 E stays at E_0, since the tension is perpendicular to $\mathbf{v}_P = \xi\dot{\phi}\,\hat{\mathbf{e}}_r$ and does no work. In cases 1 and 5 E grows. In case 1, $T = mg$ and $E = E_0 + mga\omega t$ rises linearly from −0.980 98 J to 0.000 02 J; in case 5 it rises from −0.930 98 J to −0.191 20 J, slightly faster than in case 1 because the swing adds $m\xi\dot{\phi}^2$ to the tension.

5.2 Critical Events

Case 3: The string wraps around the spool until $\xi = 0$. This happens at $t = 0.270\,947$ s, when $\phi = -5$ rad ($-286.5°$, a bit less than one full turn). The particle touches the spool at $P = (0.056\,73,\ 0.191\,78)$ m.

$\dot{\phi}$ grows without limit. With $\omega = 0$ the energy is conserved, so the speed of P stays finite (it is $v = 8.75$ m/s at contact) and, since $v = \xi|\dot{\phi}|$, we get $|\dot{\phi}| = v/\xi$. When ξ goes to zero, $|\dot{\phi}|$ goes to infinity: at the last step ($\xi = 10^{-6}$ m) it is already $\dot{\phi} = -8.8\times10^{6}$ rad/s. The tension grows in the same way ($T = 7.7\times10^{6}$ N at that point). In a real case the particle would have some size and would hit the spool earlier.

Case 4: The tension is $T = m\xi\dot{\phi}^2 + mg\cos\phi$. The first term is always positive, but the second one is negative when $|\phi| > \pi/2$, that is, when the string points above the horizontal through B. So T can only reach zero there. On the right swing the maximum angle is $\phi = 1.266\,67$ rad $< \pi/2$, so the string stays tense. On the left swing $|\phi|$ passes $\pi/2$ and the tension vanishes at $t = 1.693\,47$ s, with $\phi = -1.585\,90$ rad and $P = (-0.685\,76,\ -0.189\,66)$ m. There P moves slowly (0.318 m/s), so $m\xi\dot{\phi}^2$ is too small to balance gravity: $\xi\dot{\phi}^2 = 0.148\,\text{m/s}^2 = g|\cos\phi|$.

Case 5 (Stop condition): The tension never reaches zero (its minimum is 0.46 N), so the integration stops because P hits the spool ($\xi = 0$) at $t = 35.509\,42$ s, at $P = (0.024\,29,\ -0.198\,52)$ m. From $\xi = \xi_0 - a(\omega t - \phi)$, this happens when $\omega t - \phi = \xi_0/a = 5$ rad. The rotation of the spool gives $\omega t = 3.551$ rad and the swing to the left gives $\phi = -1.449$ rad, which adds up to 5 rad. In case 1 only the spool winds the string, so it takes 50 s; here the swings help and contact comes 14.5 s earlier.

6 Motion After the Loss of Tension

After t_s the string cannot push, Eq. (1.1) no longer holds and P has two degrees of freedom again, so `diffeq` is no longer valid. To keep integrating case 4, the code would need these changes:

1. **A second model.** When `stopfun` detects $T = 0$, the integration switches to free flight under gravity, $\ddot{x}_P = 0$ and $\ddot{y}_P = -g$, with a new state vector $\mathbf{Y} = [x_P, y_P, \dot{x}_P, \dot{y}_P]^\mathsf{T}$ and a new function for $d\mathbf{Y}/dt$.
2. **Initial conditions at the switch.** Position and velocity are continuous at t_s, so the flight starts from $\mathbf{r}_P(t_s)$ and $\mathbf{v}_P(t_s)$ given by Eqs. (1.9) and (1.10).
3. **A new event function.** During the flight the string is slack while P is closer to the spool than the string allows. With $\omega = 0$ the wrapped part does not move, so the string recovers again tension when $|\mathbf{r}_P - \mathbf{r}_B(\phi_s)| = \xi_s$. A second event, $|\mathbf{r}_P| = a$, would detect if P hits the spool.
4. **Back to the constrained model.** When the string becomes tensed, it pulls P with an impulse that removes the velocity component along the string. The integration with `diffeq` and `stopfun` restarts from the new ϕ and $\dot{\phi} = \mathbf{v}_P \cdot \hat{\mathbf{e}}_r/\xi$, and the two models alternate in a loop until the final time.

7 Conclusions

- The particle has one degree of freedom, ϕ. This means that using the constraint $\xi = \xi_0 - a(\omega t - \phi)$, the motion of the problem can be described with only the unknown ξ and its derivatives choosing the right reference frame. Applying Newton's second law with the acceleration in the chosen reference frame, a rotating reference frame centered at the center of the spool and rotating with angular velocity $\dot{\phi}$, gives a tension-free equation for $\ddot{\phi}$ and an explicit $T(t, \phi, \dot{\phi})$.
- As we have derived in section 5.1, $\dot{E} = Ta\omega$. This means that energy is conserved only with the spool at rest (cases 2–4) and grows whenever the spool winds the string in (cases 1 and 5). This is due to the fact that we are not taking into account the force that makes the spool spin with constant angular velocity that must be doing the work that is added into the system.
- We have learned why we have to use a stop function that takes into account different reasons to stop the numerical solution of the differential equations of the motion of P. For example, in case 3, the particle hits the spool at $t = 0.270\,95$ s, at $(0.056\,73, 0.191\,78)$ m, and $\dot{\phi}$ grows like $1/\xi$, which would make it approach infinite velocity. In case 4 the string losses tension at $t = 1.693\,47$ s, after BP rises above the horizontal through B. Case 5 stops by contact with the spool at $t = 35.509\,42$ s, which would once again cause the velocity of P to approach infinity.

- As discussed in section 6, if we follow case 4 after the loss of tension ($T = 0$) we need a second, free-flight model and an event that detects when the string becomes tensed if we want to know the motion of the system after this has happened.
- In this experiment we have visualized how the different initial conditions of a problem affect its outcome. We have seen that despite only changing 2 parameters in 5 different cases and leaving all the other equal and the system setup the same, the simulations have led to different results. For example, mechanical energy is not conserved in cases 1 and 5 without taking into account the work that the force that is making the spool spin is doing. Also, the trajectory described by P has changed: in case 1, it has simply gone up in a straight line; in case 2; it has oscillated with a constant non-wrapped string length $\xi = \xi_0$; in case 3, it has gone to the left while the string was being pulled by the spool, creating an spiral trajectory; in case 4, it has swinged to the right and then to the left while the string was being pulled until the tension has vanished and the simulation has ended; in case 5, it has oscillated while the string was being pulled up.

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